

Exercise 5: Implicit Euler backward Scheme

Unconditionally Stable

$$C^{n+1} = \frac{C^n}{(1 + k\Delta t)}$$

$$C^{n+1} = \left(\frac{1}{1 + k\Delta t} \right) C^n$$

- Isolate the multiplier acting upon the state variable C^n

defined multiplier as: $A = \frac{1}{1 + k\Delta t}$

Defining stability: ≤ 1

∴ Numerical Scheme to be stable, the multiplier must be

Satisfy the bounding envelope: $0 \leq A \leq 1$

for $k > 0$ & $\Delta t > 0$

∴ $k\Delta t > 0 \rightarrow 1 + k\Delta t > 1$

} making the denominator of the multiplier

$$A = \frac{1}{1 + k\Delta t} > 0 \quad \forall \quad \Delta t > 0$$

$$A = \frac{1}{1 + k\Delta t} < 1 \quad \forall \quad \Delta t > 0$$

$$\therefore 0 < \frac{1}{1 + k\Delta t} < 1$$

this is true for the unconditionally stable